**Math 241, F18** **Name:** Answer

**Quiz 3**

<u>Instructions:</u> **There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.**

**Problem 1.** (5 points) Find an equation of the plane through the points $(0,1,2)$, $(2,0,1)$, and $(1,2,0)$. (*Hint: Use the cross product*).

Take $P=(0,1,2)$, $Q=(2,0,1)$, $R=(1,2,0)$

$\overrightarrow{PQ}=\langle 2,-1,-1\rangle$, $\overrightarrow{PR}=\langle 1,1,-2\rangle$

$$n=\overrightarrow{PQ}\times\overrightarrow{PR}=\begin{vmatrix} i & j & k \\ 2 & -1 & -1 \\ 1 & 1 & -2 \end{vmatrix}=[(-1)(-2)-(-1)1]i-[2(-2)-(-1)1]j+[2\cdot 1-(-1)\cdot 1]k$$

$$=3i+3j+3k$$

Thus the equation of the plane is given by

$$3(x-0)+3(y-1)+3(z-2)=0$$

$$3x+3y+3z=9$$

$$\underline{x+y+z=3}$$

**Problem 2.** (5 points) The motion in space of a particle is given by $\mathbf{r}(t)=\langle 2\cos t, 3\sin t, 4t\rangle$. Compute the velocity, speed and acceleration of the particle at time $t=\pi/2$.

$r(t)=\langle 2\cos t, 3\sin t, 4t\rangle$

$\Rightarrow v(t)=r'(t)=\langle -2\sin t, 3\cos t, 4\rangle$

$a(t)=v'(t)=r''(t)$

$=\langle -2\cos t, -3\sin t, 0\rangle$

At time $t=\pi/2$

$v(\frac{\pi}{2})=\langle -2\sin\frac{\pi}{2}, 3\cos\frac{\pi}{2}, 4\rangle=\langle -2,0,4\rangle$

Speed $|v(\frac{\pi}{2})|=\sqrt{(-2)^2+0^2+4^2}=\sqrt{20}=2\sqrt{5}$

$a(\frac{\pi}{2})=\langle -2\cos\frac{\pi}{2}, -3\sin\frac{\pi}{2}, 0\rangle=\langle 0,-3,0\rangle$